

COPYRIGHT RESERVED

BCA (Sem-IV) – Design

Analysis of Algorithm (401) Core-VIII

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section–A

Objective type Questions

1. Choose the correct alternative for each of the following questions : 2×10=20

(a) First Statement : Algorithm is

S150/8/1

(1)

(Turn over)

independent from any programming languages.

Second Statement: A pseudo code describes the high level abstract process.

(i) First Statement is correct second is incorrect.

(ii) Second statement is correct; first is incorrect.

(iii) Both statements are correct.

(iv) Both statements are incorrect.

(b) There are ___ important characteristics of an algorithm can have.

(i) 3

(ii) 2

(iii) 5

(iv) 4

S150/8/1

(2)

Contd.

(c) 'read(a)' in any algorithm means :

(i) Reading the variable 'a'

(ii) Input a value and store in a variable named 'a'

(iii) Reading the value from variable 'a'.

(iv) None of those

(d) Algorithm is case sensitive ?

(i) Always

(ii) Sometimes

(iii) Never

(iv) None of these

(e) Binary Search follows the following :

(i) Divide and conquer

(ii) Greedy algorithm

(iii) Dynamic programming

(iv) None of these.

S150/8/1

(3)

(Turn over)



(f) Recursive takes _____ than iterative.

- (i) More Time
- (ii) More memory
- ✓ (iii) Both (a) & (b)
- (iv) None of these

(g) Which algorithm uses memorization ?

- ✓ (i) Dynamic Programming
- (ii) Greedy Algorithm
- (iii) Divide and conquer
- (iv) None of these

(h) _____ sort is a sorting technique that sorts the elements by first grouping the individual digits of same place value.

- (i) Heap
- (ii) Merge
- ✓ (iii) Radix
- (iv) None of these

S150/8/1

(4)

Contd.

(i) Travel salesman problem can be best solved by :

- ✓ (i) Greedy approach
- (ii) Dynamic Algorithm
- (iii) Divide and Conquer
- (iv) None of these

(j) STPSET is used in :

- (i) Kruskal's algorithm
- (ii) Dijkstra's Algorithm
- ✓ (iii) Prim's algorithm
- (iv) None of these.

Section-B

Short Answer type Questions

Answer any four questions of the following :

4×5=20

②. Differentiate algorithm and pseudo code.

S150/8/1

(5)

(Turn over)



- ③. Differentiate between Iteration and recursion technique.
4. Write an algorithm for Bubble sort.
5. Write an algorithm for Binary search.
- ⑥. Define divide and conquer. How do we implement it in solving big problems?
7. What are the features important for "correctness of an algorithm"?
8. Create a decision tree for a student registration process.
- ⑨. Explain all three steps involved in divide and conquer.

Section-C

Long Answer type Questions

Answer any two questions of the following :

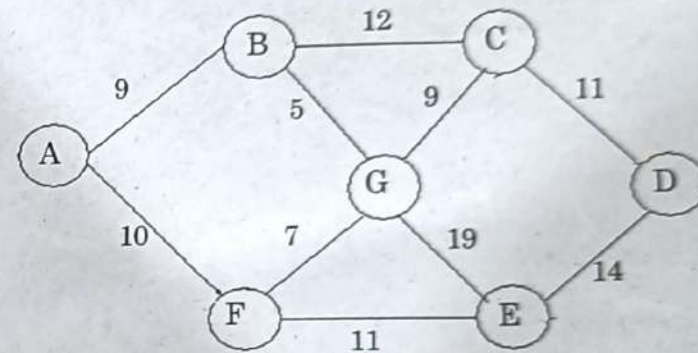
$$15 \times 2 = 30$$

S150/8/1

(6)

Contd.

10. With the help of counting coin problems, explain how dynamic technique is better than greedy approach.
11. Find out MSI using Krushkal's algorithm.



- ⑫. Through an example show the working of Dijkstra's algorithm for shortest path routing.
- ⑬. Differentiate between BFS and DFS.

— x —

S150/8/1

(7)

(P-300)